



Introduction & Installation of Power BI



**HELP 10 MILLION STUDENTS
TO LEARN THE
TECHNOLOGY IN EASY WAY**



- **Python Programming**
- **Advanced Python Programming**
- **Python Full Stack**
- **Machine Learning**
- **Artificial Intelligence**
- **Data Science**
- **Java Programming**
- **Java Full Stack**
- **React JS**
- **Web Development Full Stack**



- **Embedded Systems**
- **IoT using Raspberry Pi**
- **Computer Vision (Open CV)**
- **Image Processing - MATLAB**
- **Artificial Intelligence - MATLAB**
- **MATLAB Simulink**
- **EV Design using MATLAB**
- **Renewable Energy Design**



DATA ANALYTICS

Preprocessing, analysing, Visualizing

Data Analysis is a process of finding insights from the information

DATA



Raw data
Messy data
Unstructured data

INFORMATION



Cleaned Data
Structured data

INSIGHTS



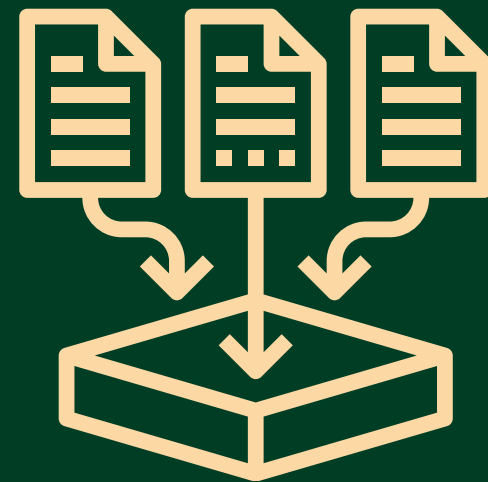
insights
data visualization



LIFE CYCLE



IDENTIFY



COLLECTION



CLEAN DATA



ANALYZE

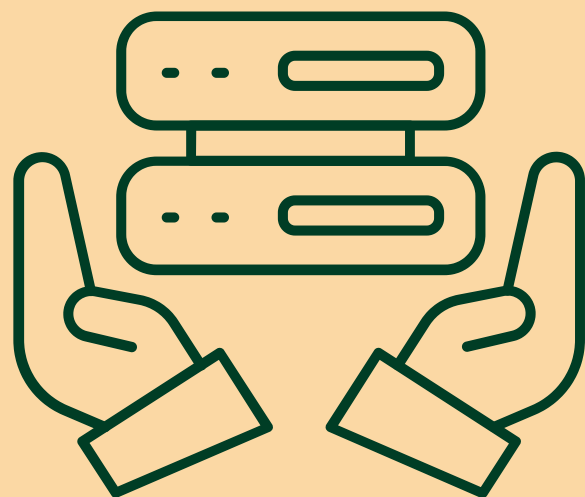


INTERPRET

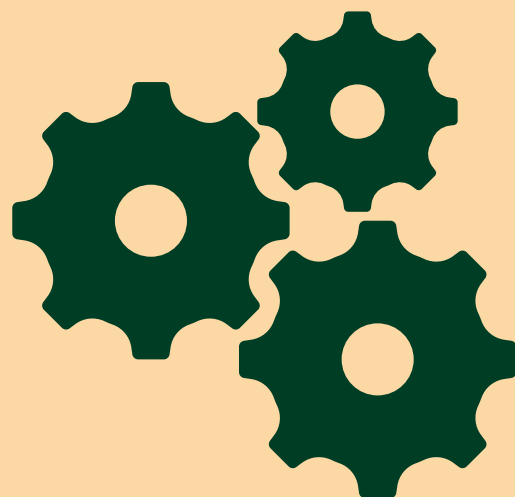


WHO IS DATA ANALYST?

A Data Analyst is a Professional who wants to work on collecting, process, and Analyzing large datasets.



DATA HANDLING



DATA MODELING



DATA REPORTING



01 PIVOT TABLE

- structuring source data
- navigating to the field list
- creating and manipulating tables,etc

02 PIVOT TABLE FORMATTING

- conditionally formatting values
- Applying Defaults and customs styles
- Adjusting table layouts etc

03 SORTING,FILTERING,GROUPING

- sorting data ,applying label and value filter
- applying wild cards
- grouping values

04 CALCULATED FIELD AND VALUES

- creating calculated fields and items
- summarizing value
- applying custom calculation etc

05 PIVOTCHARTS

- building pivot charts
- adding slicer and time lines
- creating dynamic dashboards

06 CASE STUDIES

- IMDS MOVIES
- Burrio rating
- Shark attacks
- san francisco alaries
- base ball

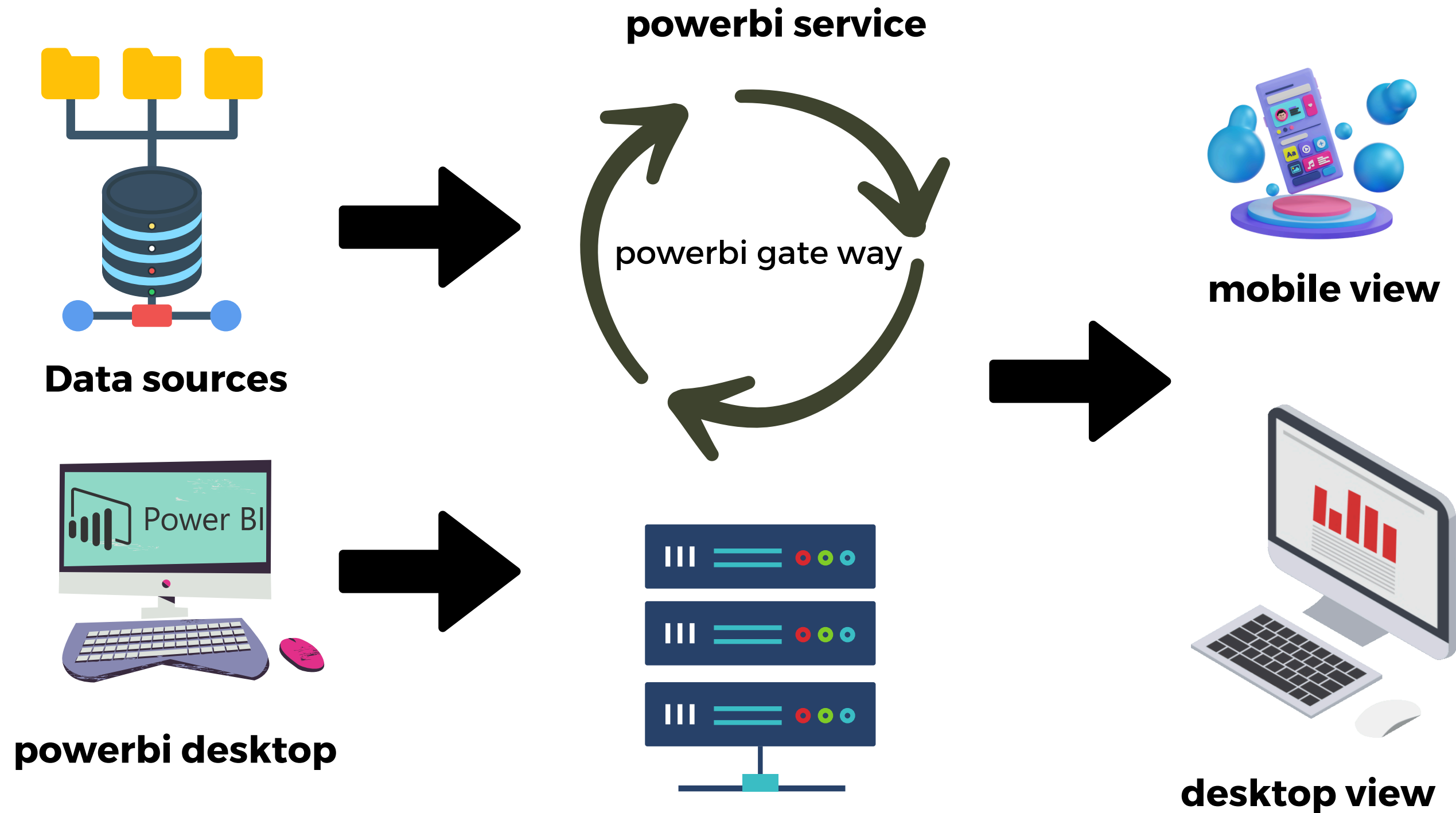


Introduction to Power BI

WHAT IS POWERBI?

- Business analysis service provided by microsoft that lets you visualize your data and share insights
- It converts data into interactive mode and usefull insights , visualize the data.
- It allows you to view , analyse and visualize huge quantities of data that cannot opened in excel
- PowerBI allows you to get insight from data and turn insights into actions to take data driven business decision

Architecture of Power BI



Power BI Components

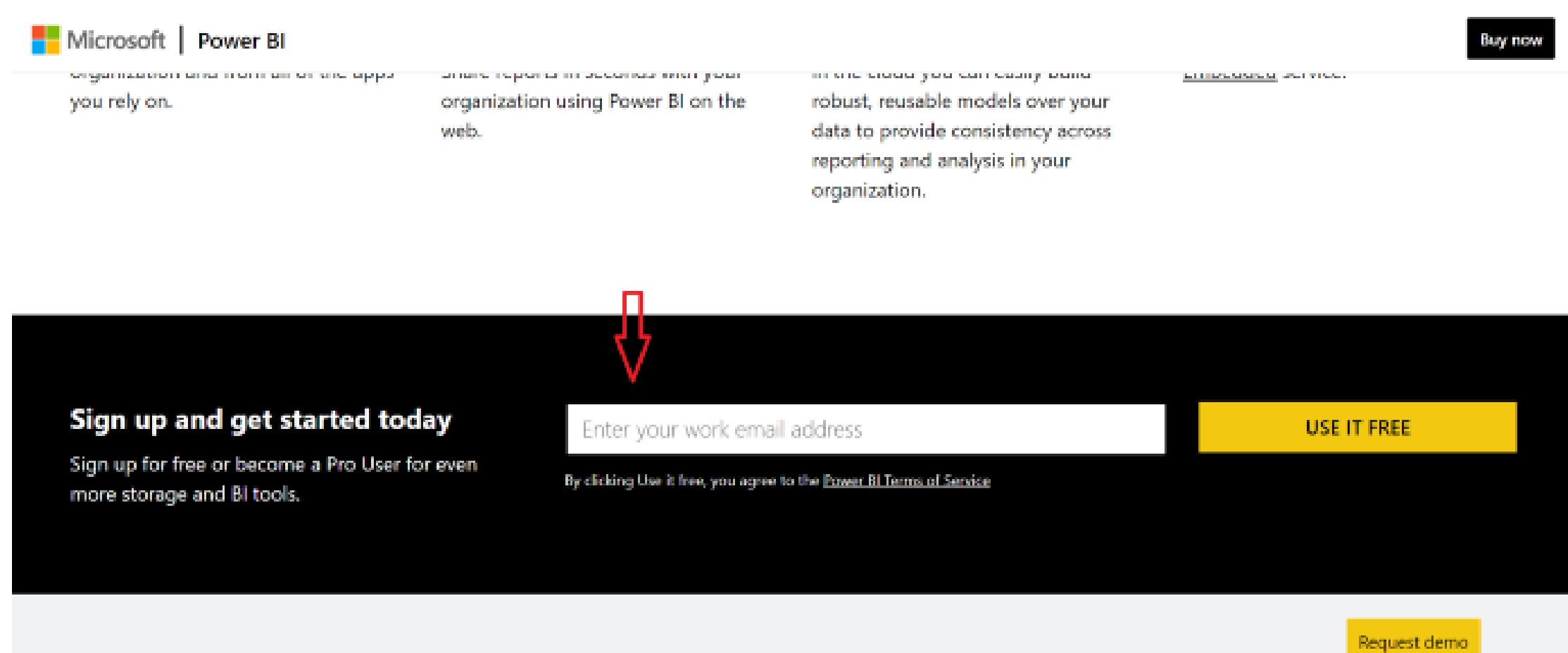
- **Power BI Desktop:** A free desktop application for Windows that allows users to create and publish reports and data models.
- **Power BI Service:** The online, cloud-based service for sharing and collaboration.
- **Power BI Mobile:** Mobile applications for iOS and Android for accessing reports and dashboards on the go.
- **Power BI Embedded:** A service that allows developers to embed Power BI reports and dashboards into their own web applications.
- **Power BI Gateway:** A tool that ensures data is transferred to the cloud securely.

Power BI Versions

- **Power BI Free:** Available for individual use. It includes basic features like creating reports and dashboards, but has limited sharing capabilities.
- **Power BI Pro:** A paid version for collaboration and sharing, which includes additional features such as the ability to share content and collaborate with others.
- **Power BI Premium:** A more advanced version, designed for larger organizations, offering dedicated cloud resources, better performance, and advanced data models.

Installation of Power BI

After installing Power BI Desktop, you may now sign in to your account on the [Power BI service online](#). In case of a new user, scroll down and sign up for free with your student or work email address.



Microsoft | Power BI

Buy now

organization and from all of the apps you rely on.

Share reports in seconds with your organization using Power BI on the web.

In the cloud, you can easily build robust, reusable models over your data to provide consistency across reporting and analysis in your organization.

[Embedded services](#)

Sign up and get started today

Sign up for free or become a Pro User for even more storage and BI tools.

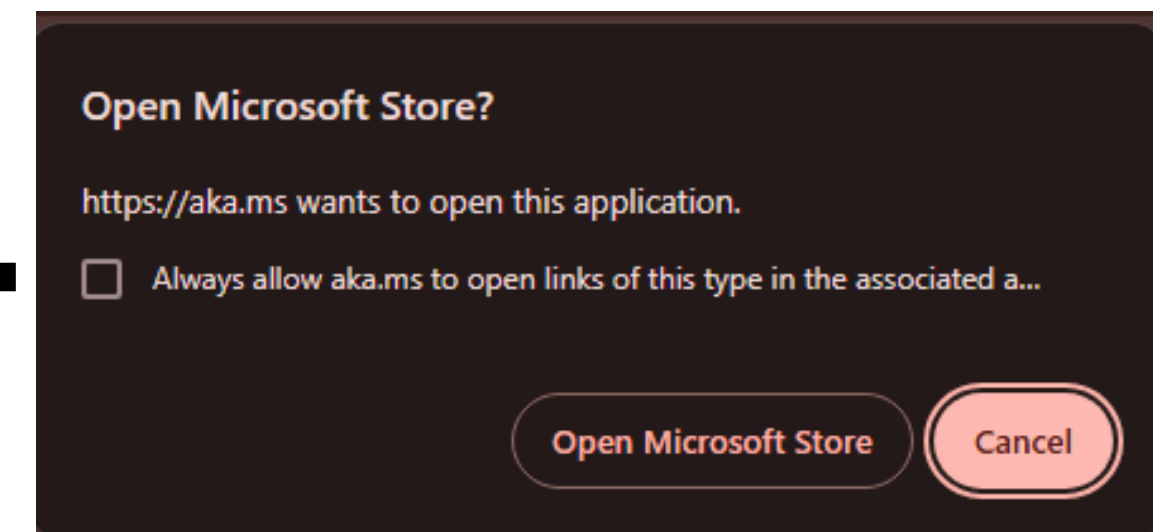
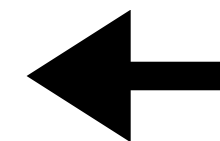
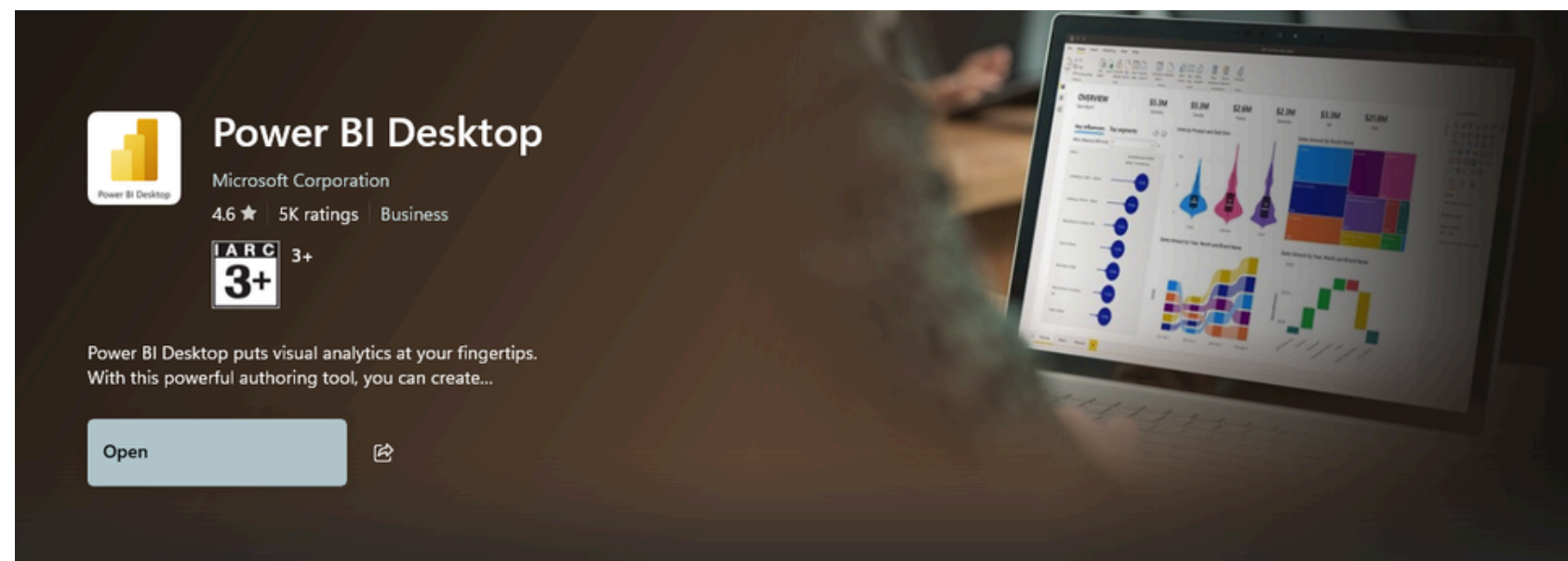
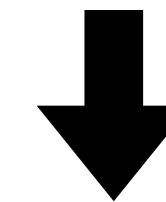
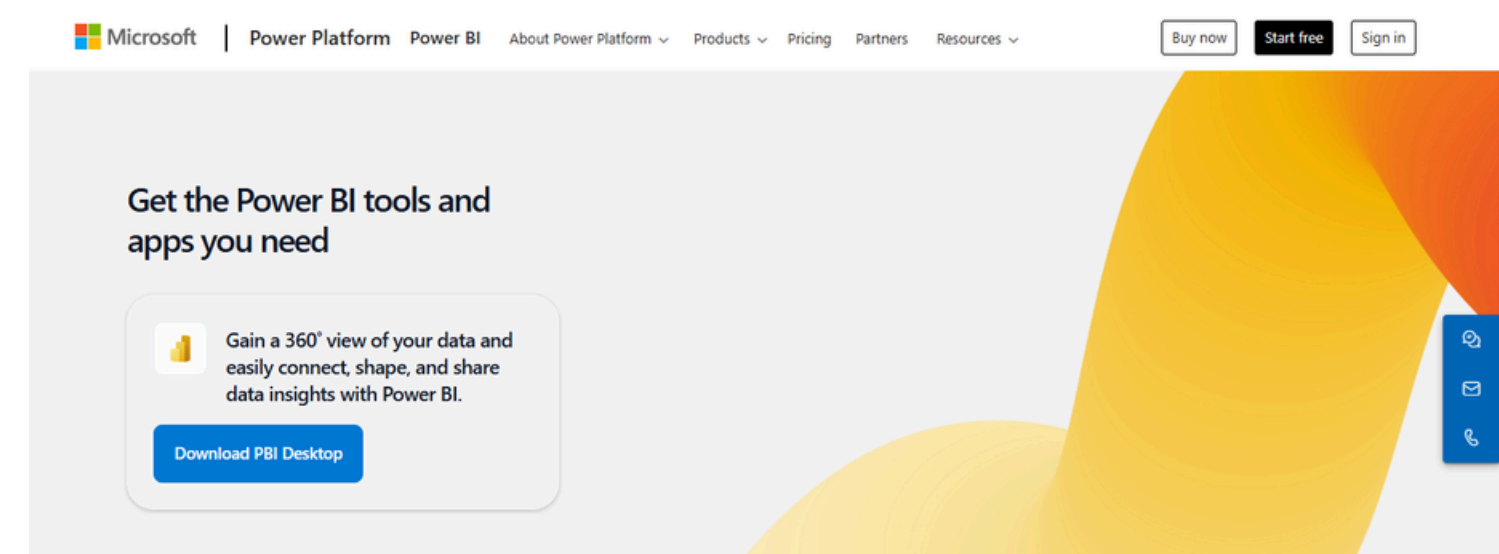
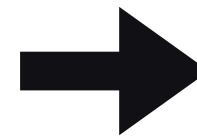
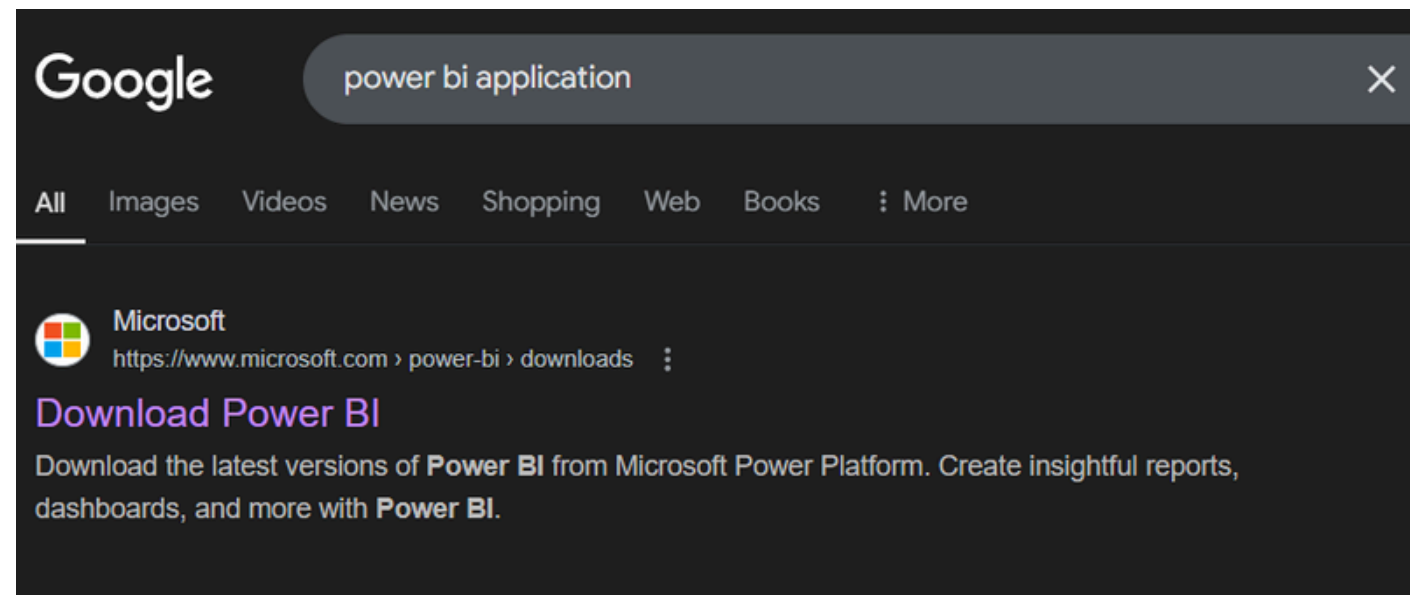
Enter your work email address

USE IT FREE

By clicking Use it free, you agree to the [Power BI Terms of Service](#)

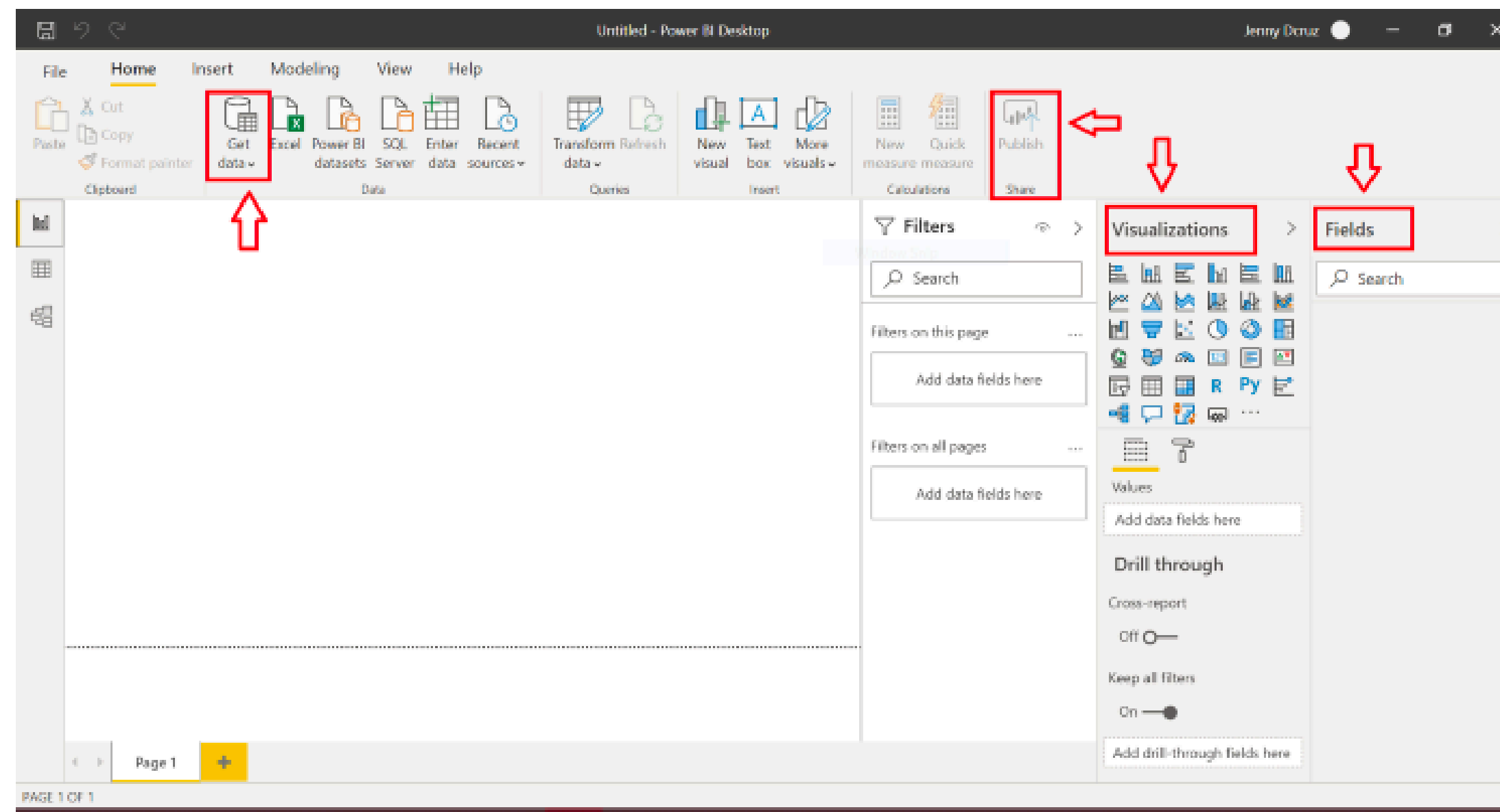
Request demo

Installation of Power BI



Installation of Power BI

Let's run Power BI Desktop after installed and study the application's primary components.



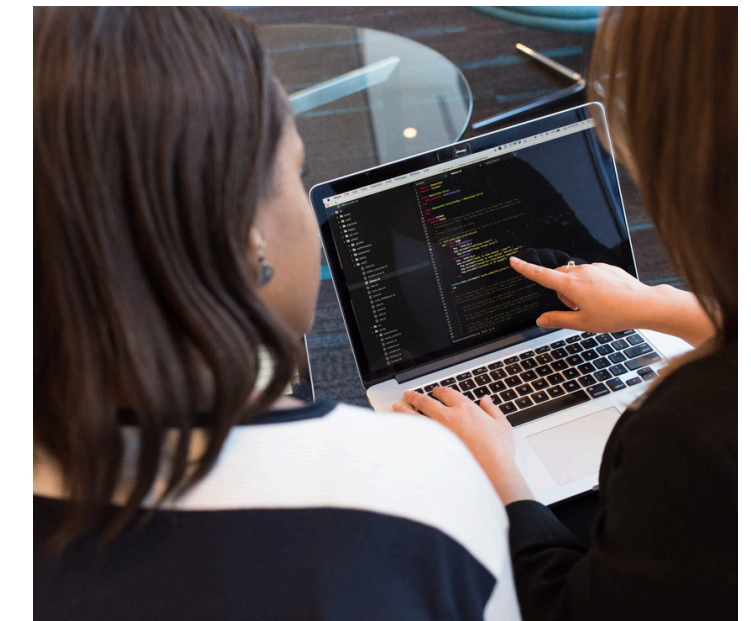
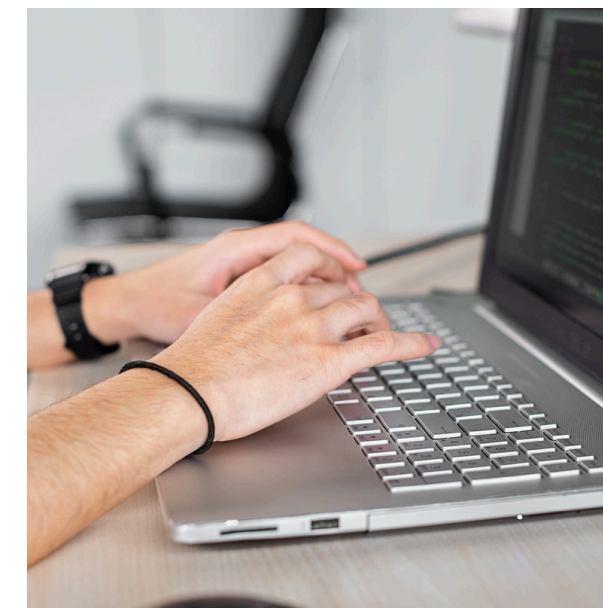
What is Python?

- It was created by Guido van Rossum, and released in 1991.
- It emphasis on code readability, shorter codes, ease of writing.
- Programmers can express logical concepts in fewer lines of code in comparison to languages such as C++ or Java.
- Python is easy to learn.
- Its syntax is easy and code is very readable.
- lot of applications in web applications, data science, rapid application development, and so on.
- Allows you to write programs in fewer lines of code

Python is a popular programming language

What can Python Do?

- 01** can be used on a server to create a web application
- 02** can be used alongside software to create work flow
- 03** can connect to database system
- 04** can be used to bigdata and complex mathematics



Hello Everyone



```
C++  
  
#include <iostream>  
  
int main(){  
    std ::cout<<"Hello everyone!";  
    return 0;  
}
```

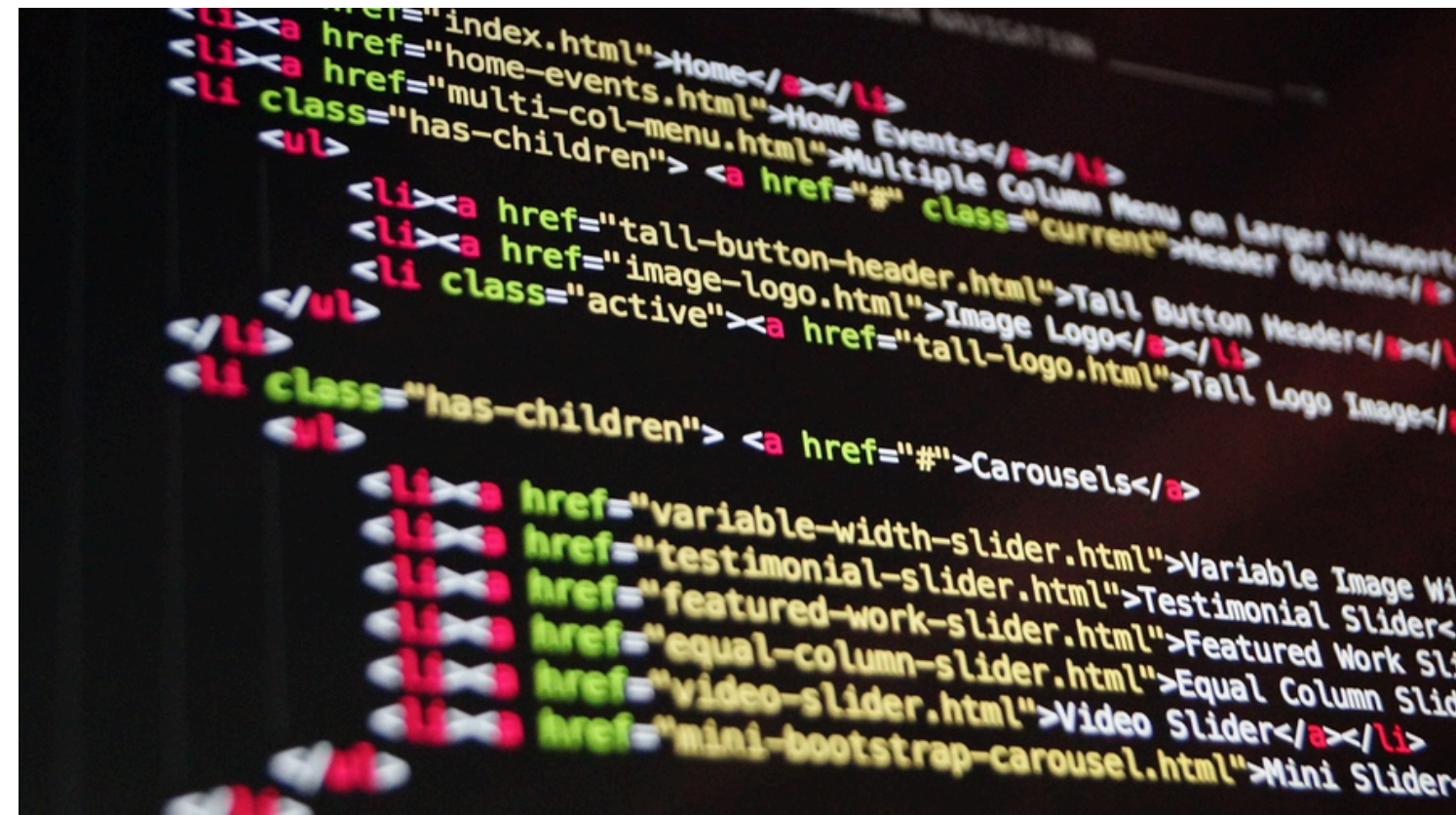
```
java  
  
class Hello everyone{  
    public static void main([]args) {  
        system.out.println('Hello everyone!');  
        //Hello everyone!  
    }  
}
```

```
python  
  
print("Hello everyone")
```

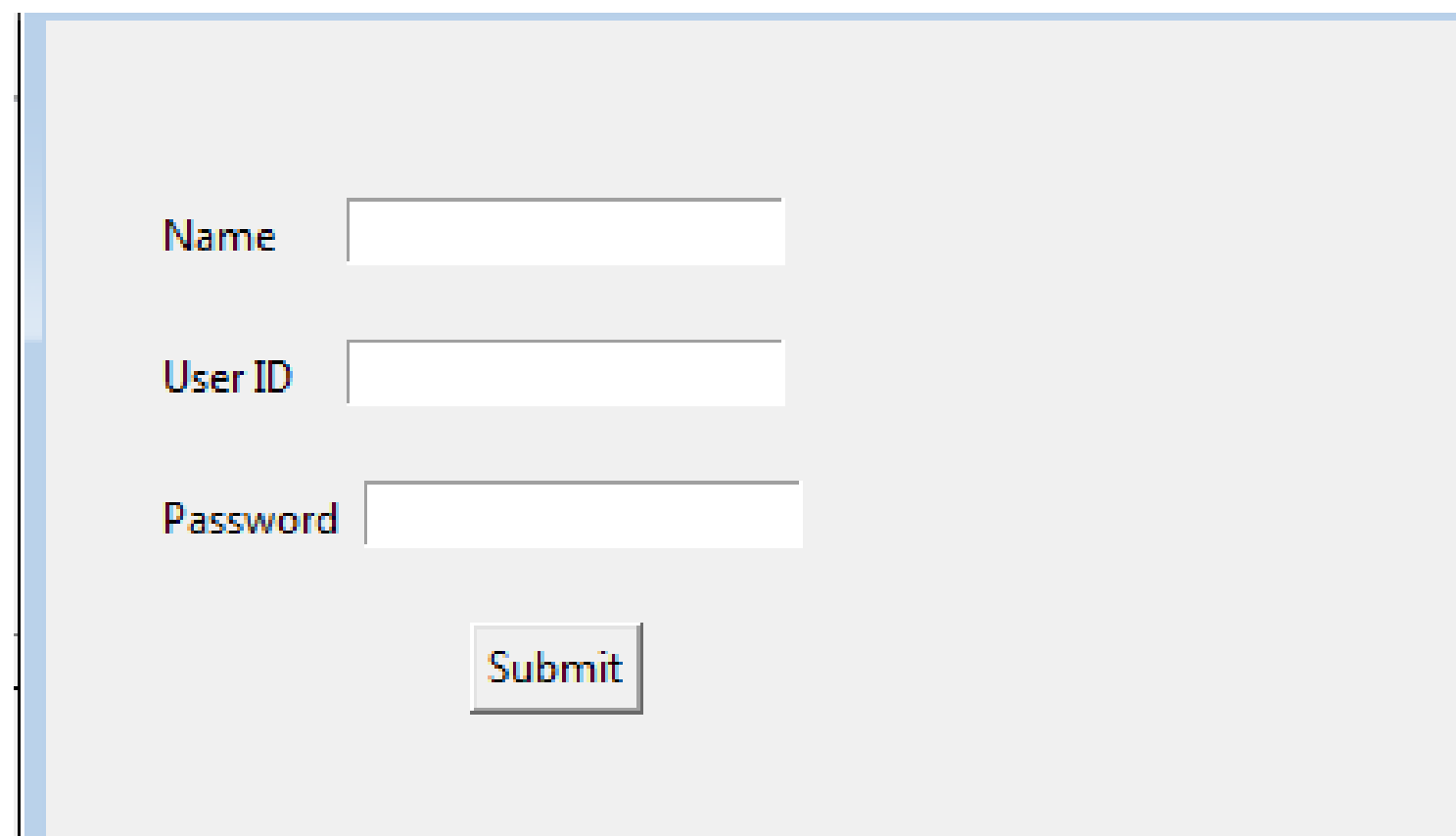
Applications of Python



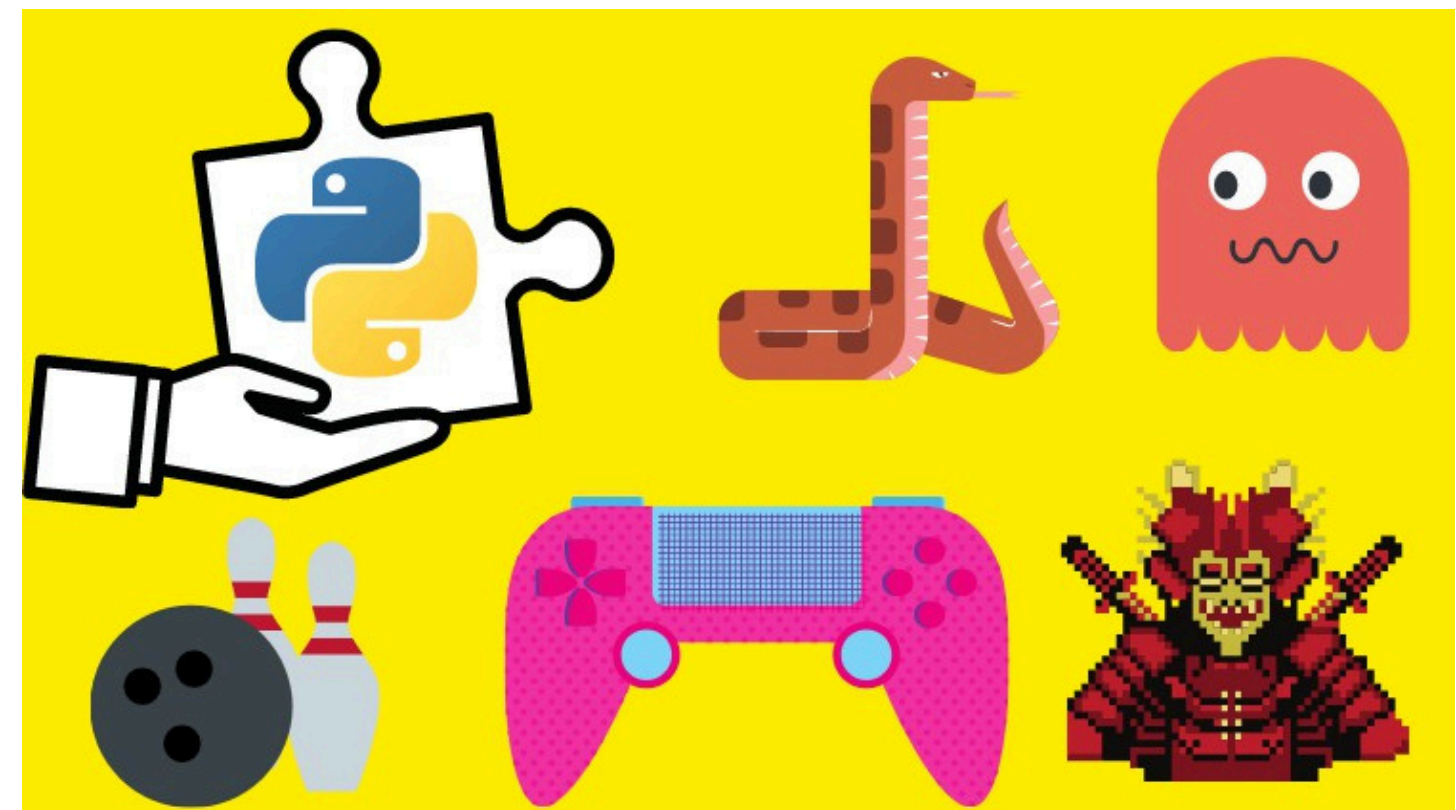
WEB APPLICATION DEVELOPMENT



DESKTOP APPLICATION

A login form with a light gray background and a blue border. It contains three input fields: "Name", "User ID", and "Password". Each field is a white rectangle with a thin gray border. Below the "Password" field is a "Submit" button, which is a gray rectangle with a thin gray border and the word "Submit" in a black, sans-serif font.

GAME DEVELOPMENT



DATA BASE



DATA SCIENCE

 **TensorFlow**

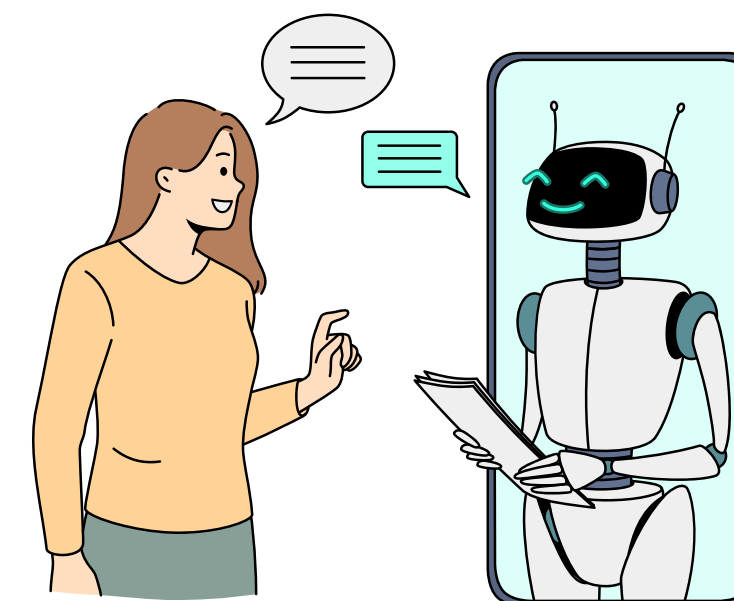
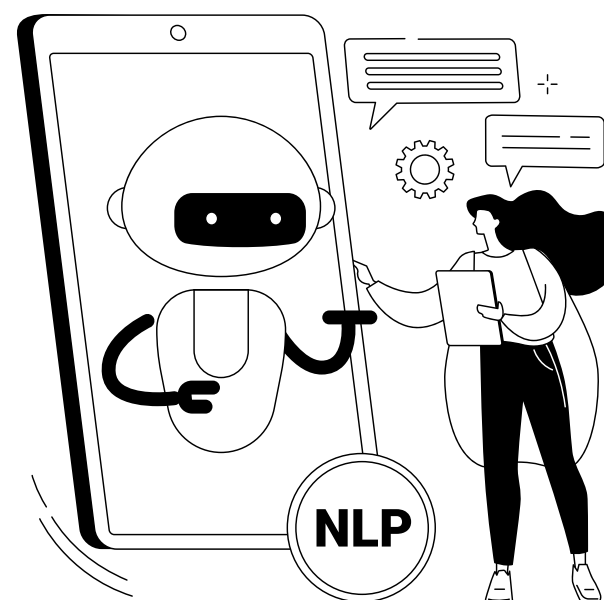
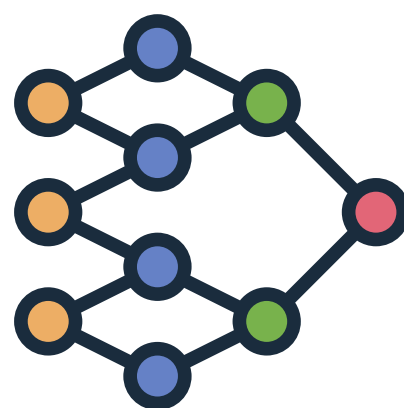
 **scikit
learn**



IMAGE PROCESSING



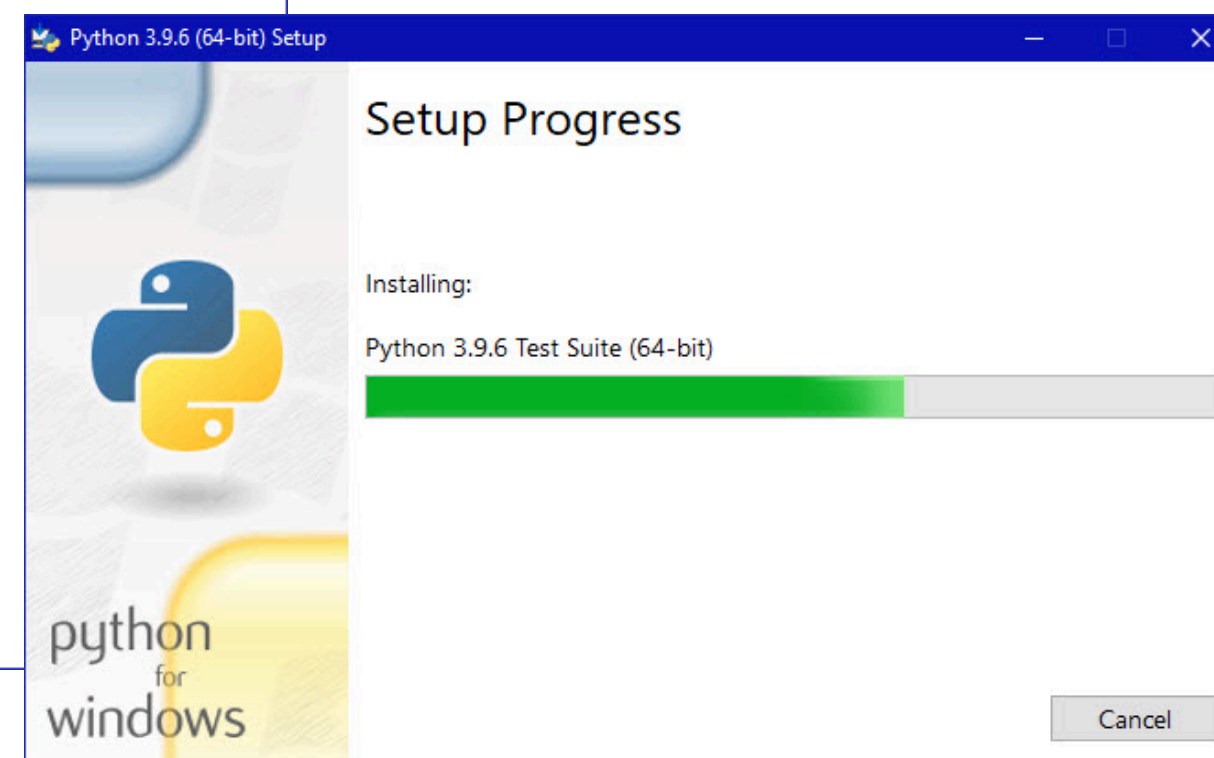
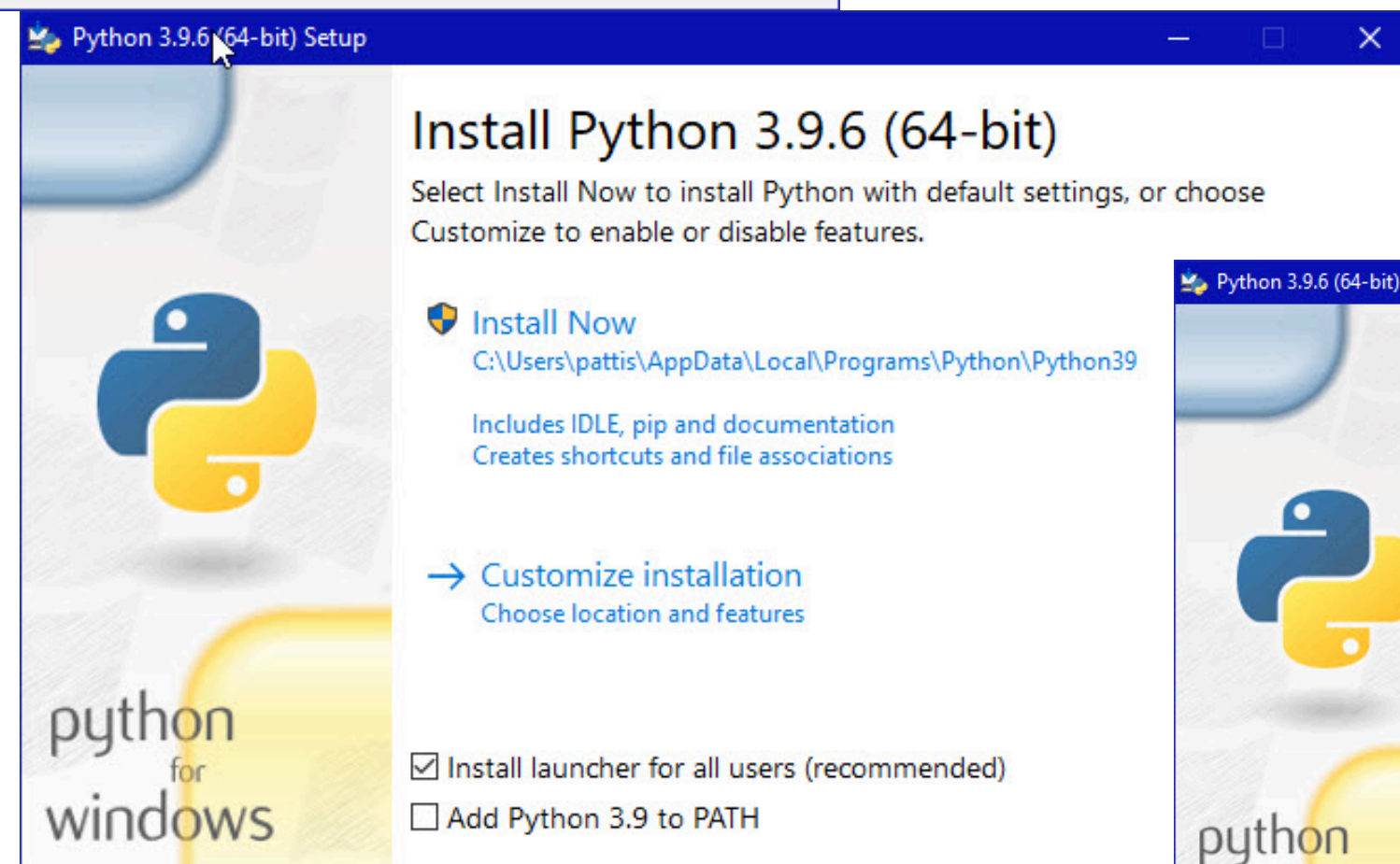
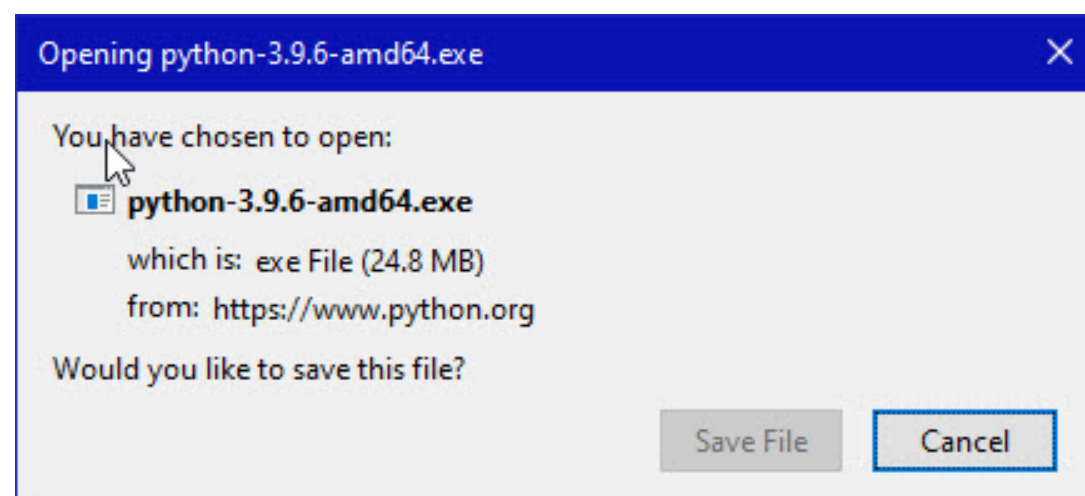
ARTIFICIAL INTELLIGENCE



PYTHON INSTALLATION

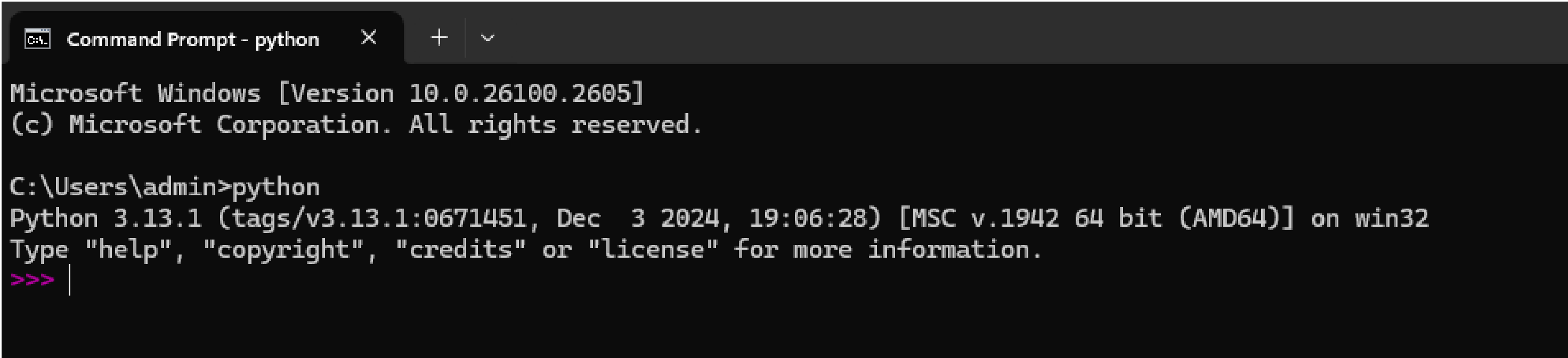


- DOWNLOAD PYTHON FROM [HTTP://WWW.PYTHON.ORG](http://www.python.org)
- INSTALL PYTHON.
- RUN IDLE FROM THE START MENU.



For the confirmation of the python installation procedure means , follow this steps:

- 1.Open Command prompt in your system.
- 2.Type this word "python"
3. you can get it your output like this image.

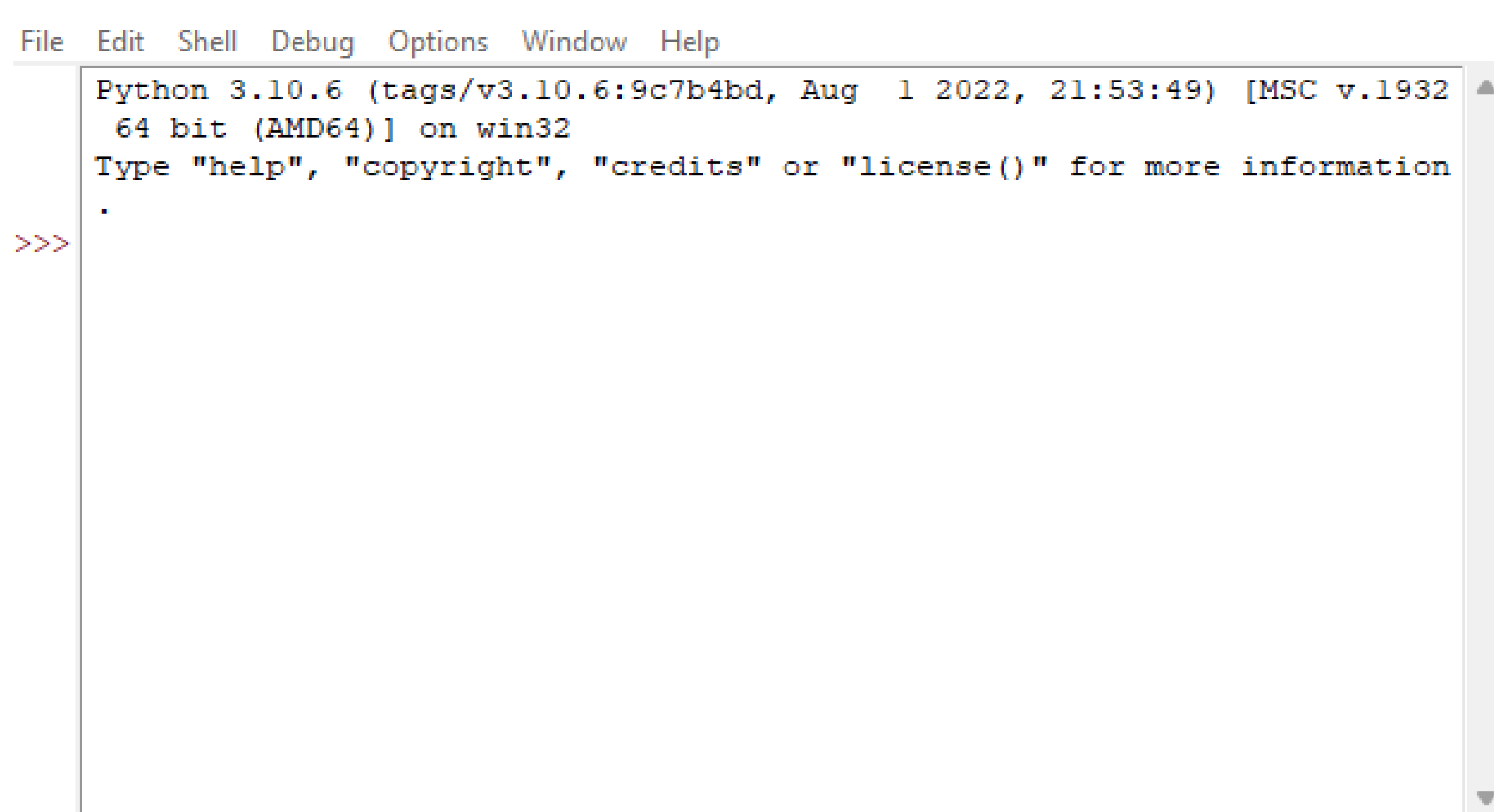


```
Command Prompt - python
Microsoft Windows [Version 10.0.26100.2605]
(c) Microsoft Corporation. All rights reserved.

C:\Users\admin>python
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> |
```

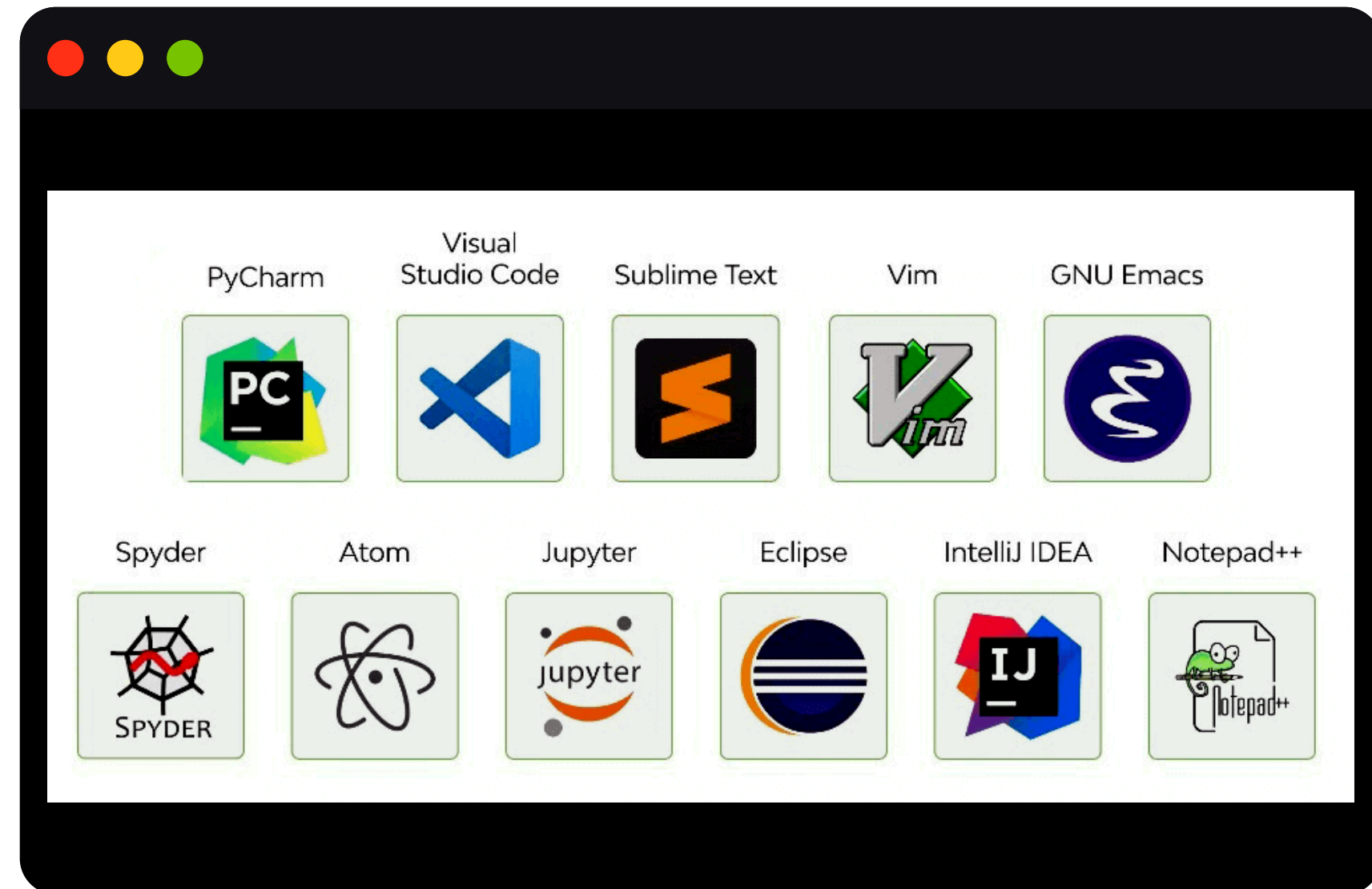
PYTHON IDLE

(INTEGRATED DEVELOPMENT AND LEARNING ENVIRONMENT)

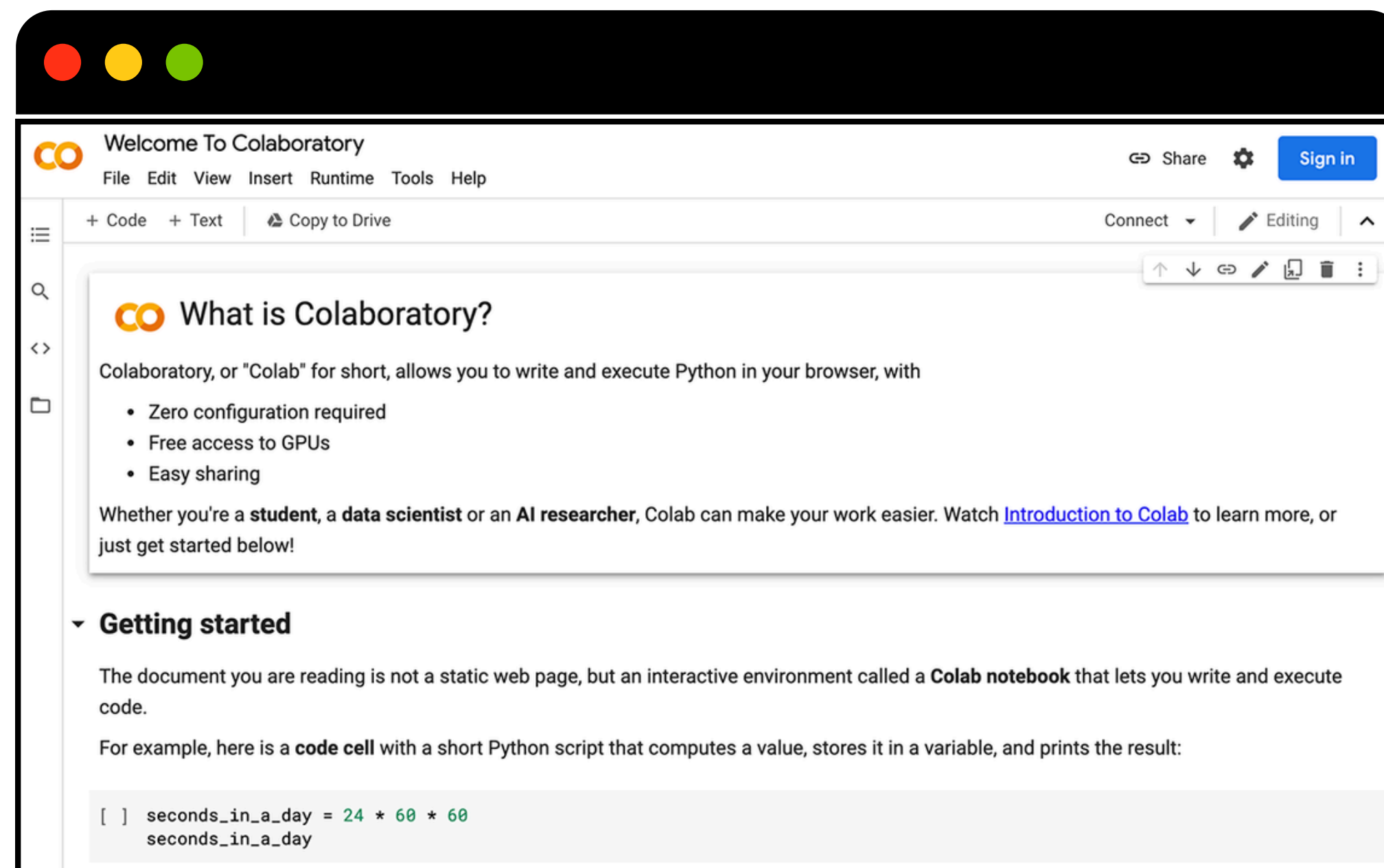


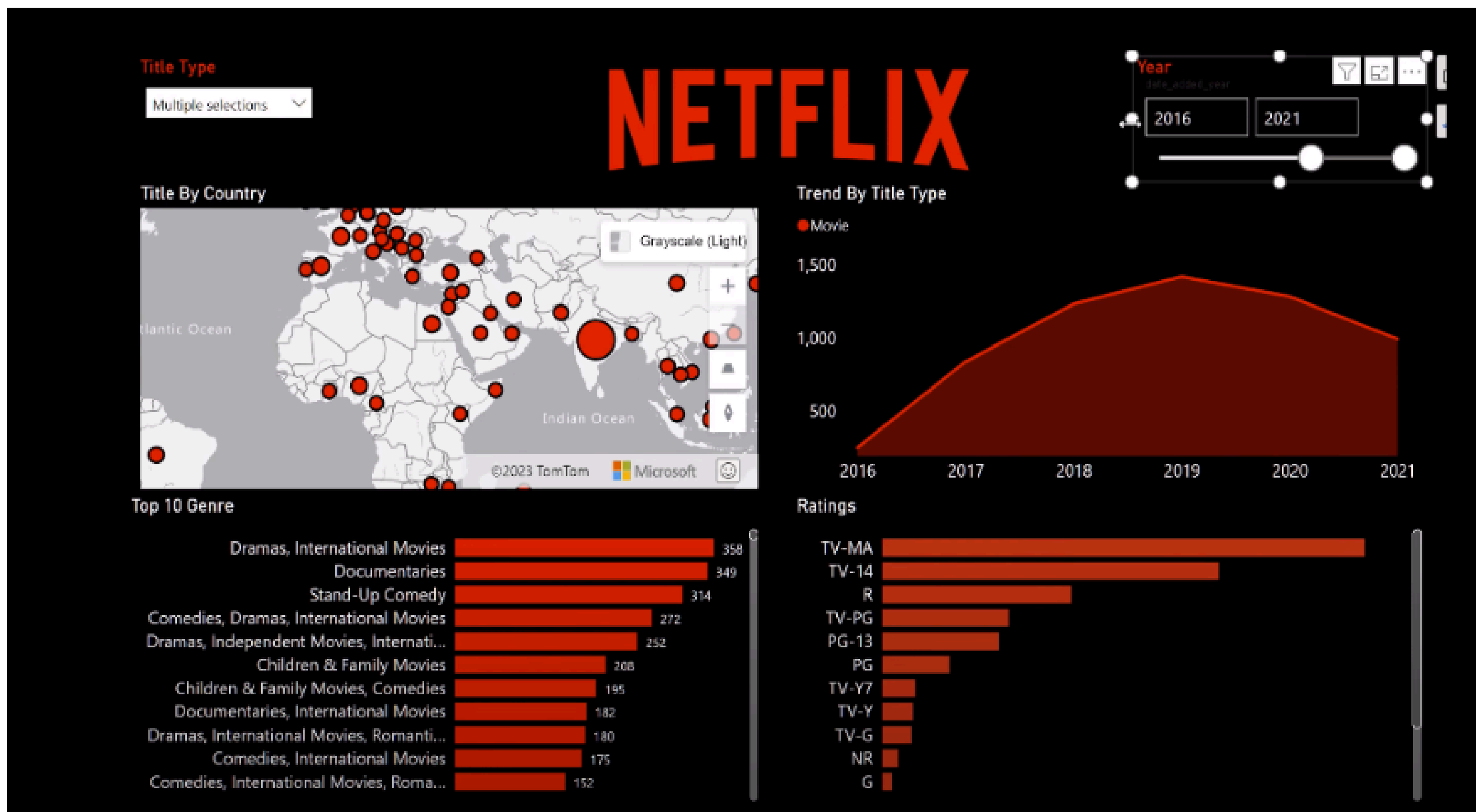
```
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932
64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information
.>>>
```

PYTHON IDE



(INTEGRATED DEVELOPMENT ENVIRONMENTS)







Python Basics

Python Basics



- **Data types**
- **Variables**
- **Expression**
- **String operations**

Data types



Data types



- Numbers
- Strings
- Boolean

Numbers



Pantech e Learning
DIGITAL LEARNING SIMPLIFIED

10

11.2

2 + 5j

10 -----> integer

11.2 -----> Float

2 + 5j -----> Complex number

Number type conversion

`int(5.32)` → 5

`float(11)` → 11.0

`complex(10)` → 10 + 0j



```
int(5.32)
```

```
float(11)
```

```
complex(10)
```

String

"40"

'Hello'

"Hello"



```
"40"
```

```
'Hello'
```

```
"Hello"
```

Boolean



Pantech e Learning
DIGITAL LEARNING SIMPLIFIED



True

False

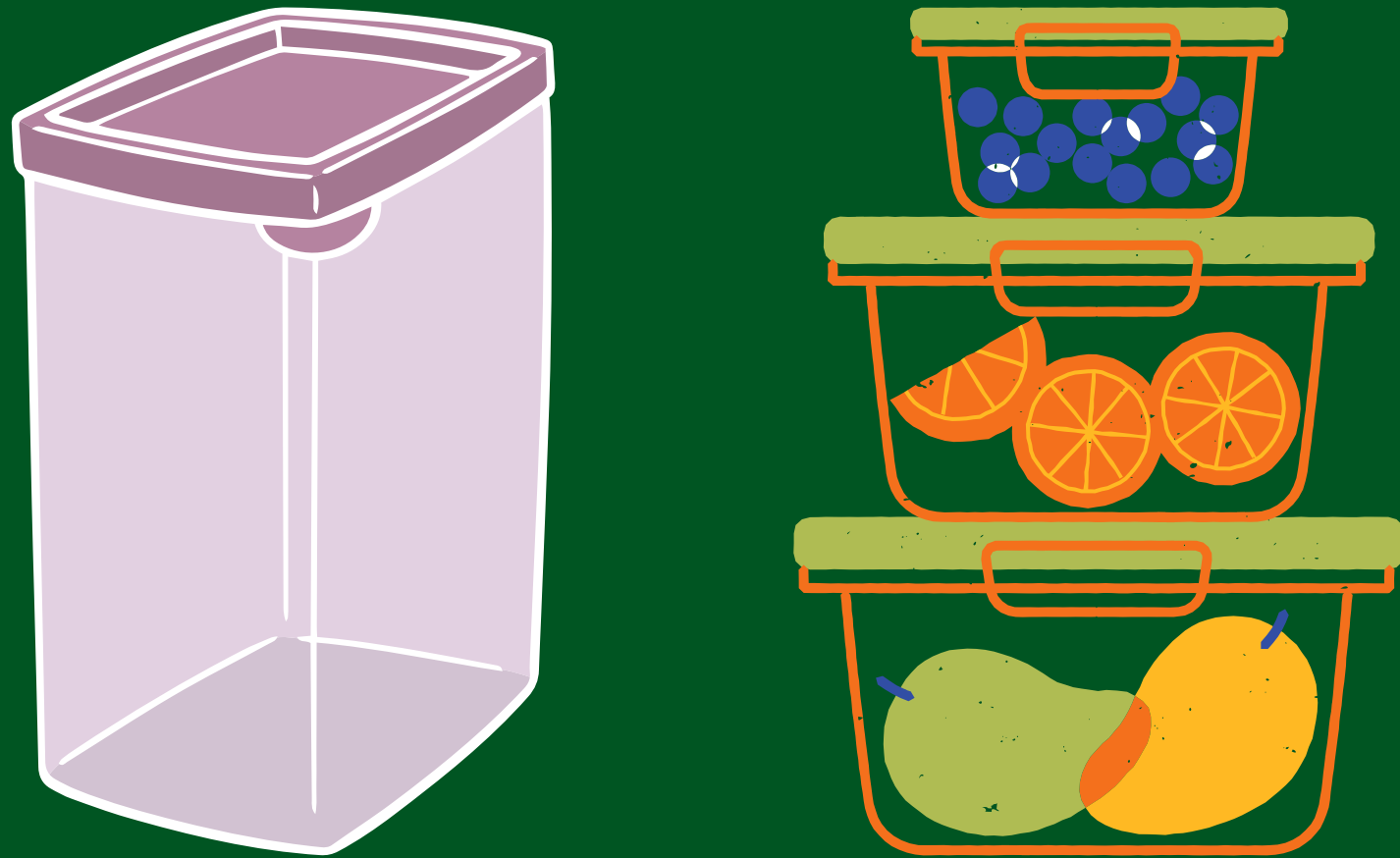
TRUE

FALSE

Variables



Variables



Variables are containers for
storing data values

variables



```
x = 5
```

```
y = "Hello"
```

```
z = True
```

- To create a variable, you simply need to assign a value to a name using the assignment operator (=)

Rules for python variable



- variable names in Python have to start with a letter or an underscore
- They cannot start with a number
- They can't contain any special characters other than an underscore
- Variable names are case-sensitive (age, Age and AGE are three different variables)

String operations



Length of a string

String →

F	A	C	E
---	---	---	---

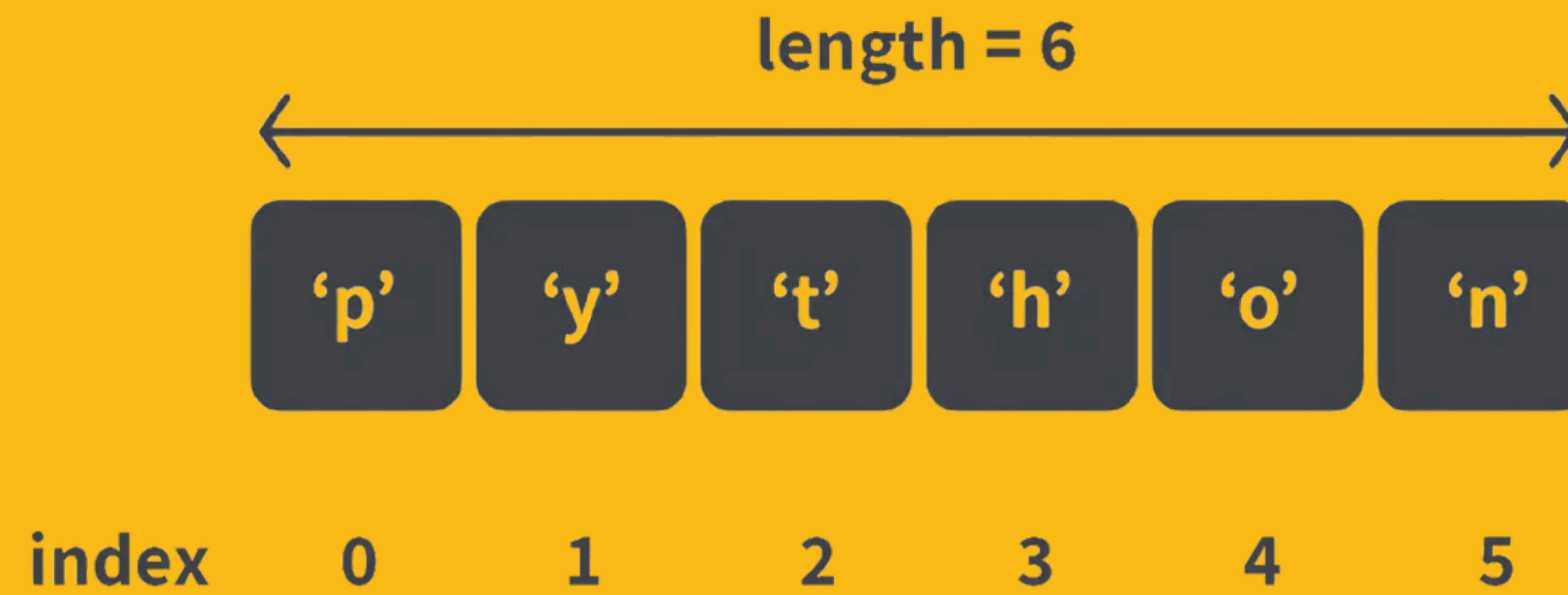
Length = 4



```
x = "Hello"
```

```
print(len(x))
```

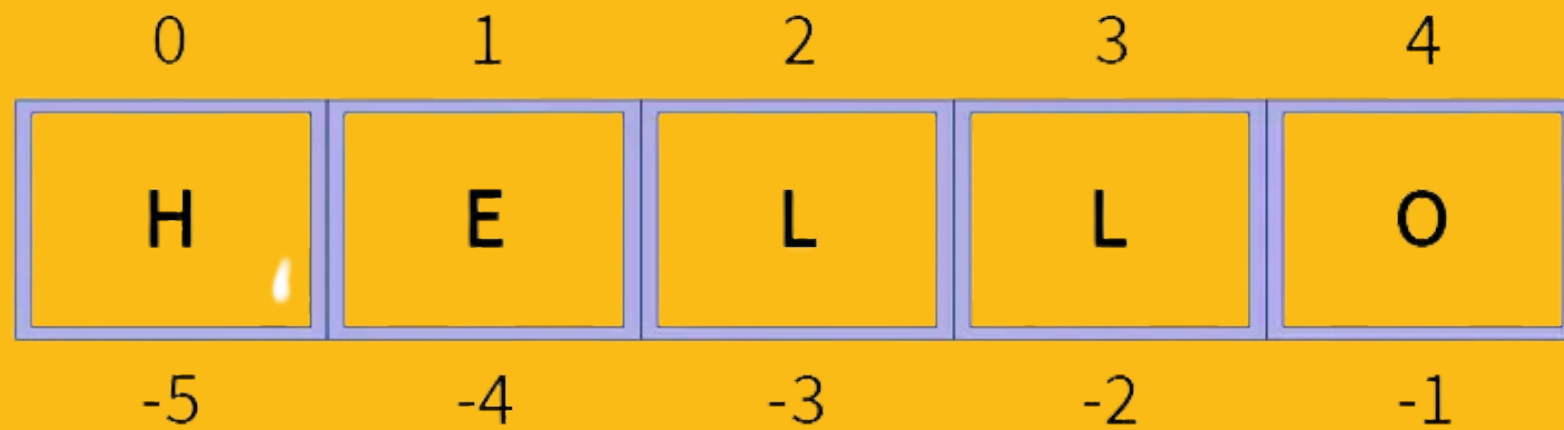
Indexing



```
x = "Hello"  
  
print(x[0])
```

string slicing

String Slicing



```
x = "Hello"

print(x[0:3])

print(x[:3])

print(x[2:])

print(x[:])
```

string concatenation

"Hello" **+** **"world"** **→** **"Hello world"**



```
x = "Hello"
```

```
y = "World"
```

```
print(x + y)
```

string format

"His age is" + 20

- we cannot combine strings and numbers
- But we can combine strings and numbers by using the format method
- placeholders should be created with {}



```
age = 20
```

```
print(f"His age is {age}")
```

Find the index



```
x = "Hello everyone, welcome to this class"
```

```
y = x.index("w")
```

```
print(y)
```

- This will return the index position of the specified string



```
x = "Hello Hello Hello Hello"
```

```
y = x.count("Hello")
```

```
print(y)
```

- This will return the number of occurrences of specified string



```
x = "hello world"
```

```
print(x.capitalize())
```

- The first character is converted to upper case, and the rest are converted to lower case



```
x = "hello world"  
  
print(x.upper())
```

- It converts all characters present in the string into upper case

```
● ● ●  
  
x = "Hello WORLD"  
  
print(x.lower())
```

- It converts all characters present in the string into lower case

Replace

```
• • •  
  
x = "Welcome to python course"  
  
y = x.replace("python", "data science")  
  
print(y)
```

- This method replaces a specified string with another specified string.

Replace



```
x = "Hello Hello Hello Hello"

y = x.replace("Hello", "Hi", 2)

print(y)
```

- The count can also be mentioned to specify how many occurrences of the old value you want to replace.



```
x = "Welcome to this course"
```

```
y = x.split()
```

```
print(y)
```

- This will split each word into separate strings



```
x = "Hi, Hello everyone, Welcome to this course"
```

```
y = x.split(", ")
```

```
print(y)
```

- You can also specify where to split



```
x = "Hello everyone, welcome to this class"
```

```
y = x.find("e")
```

```
print(y)
```

- It finds the index of first occurrence of specified string



THANK YOU

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Website : www.pantechlearning.com

